Stochastic processes.

Today: what is?

why useful?

i.i.d. sequences & processes built from these

conditioning on past & present information

the Markov property } Section 1.1-1.2 of essentials book.

transition probabilities

Definition: stochastic process: collection of

random variables on a common probability space
indexed by time (N if discrete, R if continuous)

$(X_n)_{n \in \mathbb{N}}$ discrete time

$(X_t)_{t \in \mathbb{R}_+}$ continuous time.

Technically, there is a sample space Ω such that

state space (often finite set or $\mathbb{N}$ or $\mathbb{R}$).

$$\forall n \in \mathbb{N}, \quad X_n : \Omega \to S$$

so $X_n = X_n(\omega)$, $\omega \in \Omega$

but usually we suppress "ω".

e.g. $P(X_n = 4)$ ω does not appear (in the background)

Formally, a "probability space" is a triple

$$(\Omega, \mathcal{A}, P)$$

fancy "A"

Ω set ($\omega \in \Omega$ is an "outcome").

$\mathcal{A}$ collection of subsets of Ω (events)

P probability measure

(countably additive function from $\mathcal{A} \to [0,1]$ s.t. $P(\emptyset)=0$, $P(\Omega)=1$.)

if $A_1, A_2, \ldots$ disjoint then $\sum_1 P(A_i) = P(\bigcup_i A_i)$.

ex: $X_n = 1 \quad \forall n. \qquad \left(P(X_n = 1 \ \forall n) = 1 \right).$

take $\Omega = \{a\}.$

$\mathcal{A} = \{\phi, \{a\}\}$

$P: \ P(a) = 1.$

i.i.d. sequence with distbtn function F:

$(X_n)_{n \geq 1}: \qquad P(X_n \leq x) = F(x) \qquad x \in \mathbb{R}.$

$\forall A_1, \dots, A_n \quad P(X_i \in A_i \ \forall i \leq n) = \prod_{i=1}^{n} P(X_i \in A_i) \quad (*)$

$A_i \subset \mathbb{R} \ \forall i.$

can take $\Omega = \{ (x_1, x_2, \dots) : x_i \in \mathbb{R} \}$.

$\mathcal{A}$ = we'll ignore.

P defined from F via $(*)$.

ex: two-state Markov chain $S = \{a, b\}$

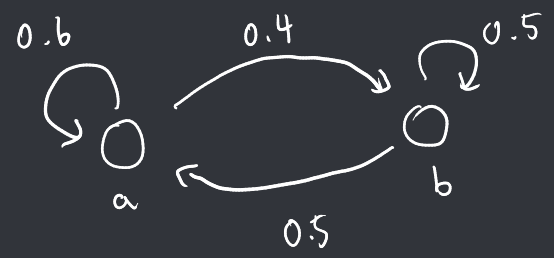


diagram represents transition probabilities.

suppose $X_0 = a$ & (X_n) has above

<u>trans. probab.</u>

$$= P(X_{n+1} = y \mid X_n = x, X_{n-1}, \ldots, X_0).$$

how to construct? Take i.i.d. sequence (u_n) of unif $[0,1]$.

If $X_n = a$ & $u_{n+1} \leq 0.6$ then $X_{n+1} = a$
else $X_{n+1} = b$

If $X_n = b$ & $u_{n+1} \leq 0.5$ then $X_{n+1} = b$
else $X_{n+1} = a$.

Has property

$$P(X_{n+1} = a \mid X_n = a, \underbrace{X_{n-1}, \ldots, X_0}_{\text{doesn't care about earlier values.}})$$

$$= P(u_{n+1} \leq 0.6) = 0.6.$$

(Markov property: holds if $\forall x, y \in S$, $\forall x_0, \ldots, x_{n-1}$,

$$P(X_{n+1} = y \mid X_n = x, X_m = x_m \ \forall m < n)$$

$$= P(X_1 = y \mid X_0 = x) \quad (*)$$

(time-homogeneous: probability does not depend on n).

denote $p(x,y) := P(X_1 = y \mid X_0 = x)$

above example has

$$\begin{bmatrix} p(a,a) & p(a,b) \\ p(b,a) & p(b,b) \end{bmatrix} = \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix}$$

transition probability matrix exists for Markov chains i.e. discrete time (X_n) with Markov property

with S finite.

easy to compute trajectory segment probabilities:

$$P(X_3 = a, X_2 = b, X_1 = a \mid X_0 = a)$$

$$= P(X_3 = a \mid X_0 = a, X_1 = a, X_2 = b)$$

$$\cdot P(X_2 = b \mid X_0 = a, X_1 = a)$$

$$\cdot P(X_1 = a \mid X_0 = a)$$

if Markov property:

$$= P(X_3 = a \mid X_2 = b) \cdot P(X_2 = b \mid X_1 = a)$$

$$P(X_1 = a \mid X_0 = a)$$

$$= p(a,a)\,p(a,b)\,p(b,a)$$

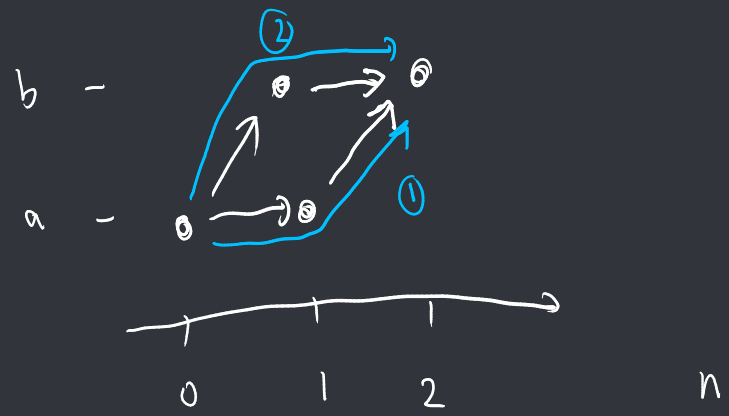
<u>ex:</u> take Markov chain w/ matrix

$$\begin{array}{c c} & \begin{array}{cc} a & b \end{array} \\ \begin{array}{c} a \\ b \end{array} & \begin{bmatrix} 1/3 & 2/3 \\ 1/2 & 1/2 \end{bmatrix} \end{array}$$

& find $P(X_2 = b \mid X_0 = a)$.

need to sum over a couple possible values of X_1.

S



path 1 : probab = $p(a,a)\,p(a,b)$

2 : " = $p(a,b)\,p(b,b)$

$$P(X_2 = b \mid X_0 = a)$$

$$= P(X_2 = b \mid X_1 = a, X_0 = a)\, P(X_1 = a \mid X_0 = a)$$

$$+ P(X_2 = b \mid X_1 = b, X_0 = a)\, P(X_1 = b \mid X_0 = a)$$

$$= p(a,b)\, p(a,a) + p(b,b)\, p(a,b) = \tfrac{2}{3} \cdot \tfrac{1}{3} + \tfrac{1}{2} \cdot \tfrac{2}{3}$$

$$= \tfrac{2}{9} + \tfrac{2}{6} = \tfrac{5}{9}$$

Things you can build from i.i.d. sequences.

- Markov chains: can generalize the above example to any finite state Markov chain (how?).

- Random walk: given i.i.d sequence (ξ_n), let

"xi" ξ

"ex-ee"

$$X_n = \xi_1 + \cdots + \xi_n.$$

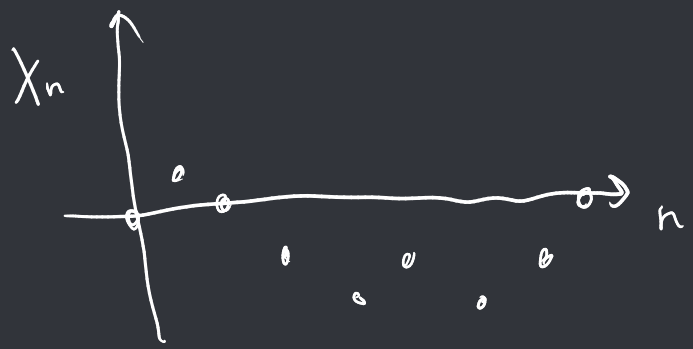
(X_n) is a random walk with increments (ξ_n).

a popular choice for ξ_n is

$$P(\xi_n = 1) = P(\xi_n = -1) = 1/2.$$

ex: flip fair coin, if H, $+1$

if T, -1.



$$E[X_n] = 0 \quad \forall n$$

$$\mathrm{Var}(X_n) = \sum_{i \le n} \mathrm{Var}(\xi_i) = \sum_{i=1}^{n} 1 = n.$$

$$X_n \overset{d}{=} 2Y_n - n, \quad Y_n \overset{d}{:=} \mathrm{bin}(n, 1/2)$$

$\overset{d}{=}$ equal in distribution.

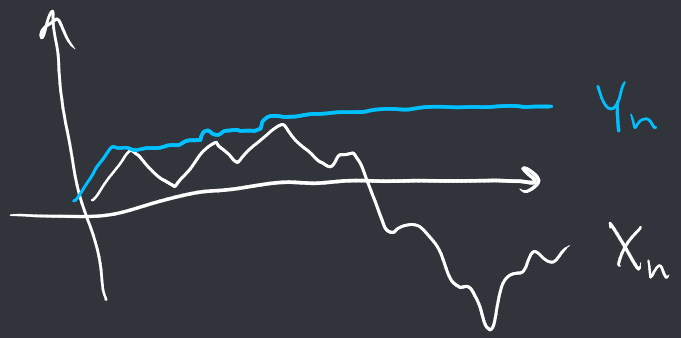
(if ξ_n i.i.d. $\xi_n \overset{d}{=} \xi_m \ \forall n, m$

but $P(\xi_n = \xi_m)$ generally < 1).

Maximum process

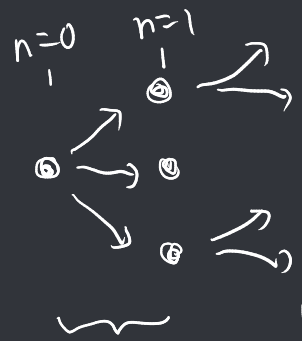
if (X_n) any stochastic process on $\mathbb{R}$,

let $Y_n = \max_{m \leq n} X_m$.



exercise: (Y_n) in general not a Markov chain, even if (X_n) is.

Branching process: model of reproduction & death.



offspring distribution

apply independently at later generations.

formally, $(\xi_{i,n})$ i.i.d. with

offspring, individ. i in generation n.

$$P(\xi_{i,n} = k) = p_k$$

$(p_k)_{k=0}^{\infty}$ offspring distbtn,

Let $Z_0 = 1$. & recursively

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{i,n}.$$

(Z_n): population size in generation n.

Question: for which (p_k) is $\underbrace{P(Z_n > 0 \ \forall n)}_{\text{long-term survival.}} > 0$?

for some intuition note

$$E[Z_n] = m^n$$

where $m = E[\xi_{i,n}]$

$$= \sum_k k p_k$$

Multistep transition probabilities

for a Markov chain w/ transition matrix p,

find formula for $p(X_n = y \mid X_0 = x)$.

saw that

$$p\Big(\underbrace{X_m = x_m \ \forall m \le n}_{\text{chance of following } (x_0, \dots, x_n) \text{ given start at } x_0.} \mid X_0 = x_0\Big)$$

$$= \prod_{m=1}^{n} p(x_{m-1}, x_m)$$

Answer: $P(X_n = y \mid X_0 = x)$ is the (x,y) entry of matrix p^n.

ex: from earlier with

$$p = \begin{bmatrix} 1/3 & 2/3 \\ 1/2 & 1/2 \end{bmatrix}$$

$$p^2 = \begin{bmatrix} 1/9 + 1/3 & 2/9 + 1/3 \\ \sim \text{exercise} \sim & \end{bmatrix}$$

$p^2(a,b)$ agrees with earlier value.

Why is it true?

Let's check that for any n, m

$$p^{n+m} = p^n p^m$$

(viewing $p^m(x,y)$ as $P(X_m = y \mid X_0 = x)$)

$$p^{n+m}(x,y) = P(X_{n+m} = y \mid X_0 = x)$$